MIL-P-46642C (MU)
21 August 1968
Superseding
MIL-P-46642B (MU)
30 June 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M92E1
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of percussion primer designated as Primer, Percussion, M92E1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder; Black.
MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

8863394 - Primer, Percussion, M92E1 Loading Assembly.
8864248 - Carton, Packing, Ammunition, for Primer Percussion, M92, for 90MM, M371E1 Cartridge.
8864249 - Box, Packing, Ammunition, for Primer, percussion, M92, for 90MM, M371E1 Cartridge.

FSC: 1390

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PUBLICATIONS

ARMY

ET-8863394- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer assembly.-The primer assembly shall comply with all requirements specified on Drawing (dwg.) 8863394 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading, shall not exceed 0.50 percent when determined as specified in 4.3.1.

3.4 Sensitivity

3.4.1 Non-functioning.-The primer, body and head assembly shall not function under an applied energy value of (2.9) inch ounces minimum (min.) when tested as specified in 4.3.2.1.

3.4.2 Functioning.-The primer, body and head assembly shall function under an applied energy value of (29.4) inch ounces maximum (max.) when tested as specified in 4.3.2.2.

3.5 Proving ground.-The primer shall function without audible hangfire (see 6.4), when tested as specified in 4.3.3.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, corrosion and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents used.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to



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represent any prior or subsequent quantities presented for

NOTES:

Varnish missing or inadequate — Visual

MINOR

None defined

593313

4.2.2.2 Assembly, Drawing F8593159, less Drawings D8593314, D8 and D8593162.

CRITICAL

| | | |
|---|---|---|
| 1. | Mixed initiators | Visual |
| 2. | Initiator not in cocked position | Visual |
| 3. | Safety pin missing or improperly engaged | Visual |

MAJOR

| | | |
|---|---|---|
| 101. | O-ring missing or not lubricated | Visual |
| 102. | Thread damaged | Visual |
| 103. | Burr | Visual |
| 104. | Foreign matter | Visual |

MINOR

None defined

B8359536.

4.2.2.3 Assembly, Drawing F8593159, less Drawings D8593314 and

CRITICAL

| | | |
|---|---|---|
| 1. | Cartridge loose | Manual |
| 2. | Cartridge retainer loose | Manual |
| 3. | Metering groove of cartridge retainer missing | Visual |
| 4. | Elongated holes of cartridge retainer missing | Visual |

and Visual
and Visual



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| | | | |
|---|---|---|---|
| Major: | AQL 0.65 percent | | |
| 101. | Primer above flush with face of Head | Gage-Visual | 11001 |
| 102. | Primer head not fully seated in tube | Visual | 11002 |
| 103. | Primer head loose in tube | Manual | 11003 |
| Minor: | AQL 1.5 percent | | |
| 201. | Length | Gage | 11004 |
| 202. | Marking missing, incorrect or unidentifiable | Visual | 11005 |
| 203. | Evidence of poor workmanship (see 3.6) | Visual | 11006 |

4.2.2.2 Primer percussion loading assembly, prior to inserting wad (see dwg. 88633394).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Charge weight | Balance | 03001 |
| Minor: | AQL 1.0 percent | | |
| 201. | Foreign matter in charge | Visual | 03002 |
| 202. | Evidence of poor workmanship (see 3.6) | Visual | 03003 |

4.2.2.3 Primer percussion loading assembly (see dwg. 8863394).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Cap loose or missing | Visual-Manual | 05001 |
| 102. | Assembly fails to freely enter and withdraw from tube gage | Gage | 05005 |



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| Minor: | | AQL 2.50 percent | |
|---|---|---|---|
| 201. | Diameter at cap (max.) | Gage | 05002 |
| 202. | Diameter at flange (max.) | Gage | 05003 |
| 203. | Length (max.) | Gage | 05004 |
| 204. | Thread damaged | Visual | 05006 |
| 205. | Evidence of poor workmanship (see 3.6) | Visual | 05007 |

4.2.2.4 Carton, prior to sealing (see dwg. 8864248).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | None defined | | |
| Minor: | AQL 1.50 percent | | |
| 201. | Any packing material missing | Visual | 06001 |
| 202. | Separator damaged to extent that primers touch each other | Visual | 06002 |
| 203. | Primer missing | Visual | 06003 |

4.2.2.5 Carton, sealed (see dwg. 8864248).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | None defined | | |
| Minor: | AQL 2.50 percent | | |
| 201. | Sealing strip fails to completely seal carton | Visual | 07001 |
| 202. | Carton torn, cut or punctured | Visual | 07002 |
| 203. | Marking missing, incorrect or unidentifiable | Visual | 07003 |
| 204. | Head end of carton not identifiable | Visual | 07004 |
| 205. | Evidence of poor workmanship (see 3.6) | Visual | 07005 |

4.2.2.6 Sealed barrier bag (see dwg. 8864248).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |



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| | | | |
|---|---|---|---|
| Major: | AQL 0.40 percent | | |
| 101. | Seal improper or incomplete | Visual | 08001 |
| 102. | Bag punctured, torn or cut | Visual | 08002 |

Minor: None defined

4.2.2.7 Packing box, prior to sealing (see dwg. 8864249).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.40 percent | | |
| 101. | Carton missing | Visual | 09001 |
| Minor: | None defined | | |

4.2.2.8 Packing box, sealed (see dwg. 8864249).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined | | |
| Major: | AQL 0.65 percent | | |
| 101. | Box broken, board loose or nail protruding | Visual | 10001 |
| 102. | Strapping missing, broken or loose | Visual-Manual | 10002 |
| 103. | Cover improperly assembled | Visual | 10003 |
| 104. | ICC nomenclature missing | Visual | 10004 |
| Minor: | AQL 0.65 percent | | |
| 201. | Marking missing, incorrect or unidentifiable | Visual | 10005 |

4.2.3 Testing.

4.2.3.1 Moisture content of black powder (see 3.3 Major defect - Code No. 03004.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test for verification at least one completed cartridge assembly from each eight hours production. Composite samples shall not be used in determining compliance. The black powder shall be tested as specified in 4.3.1 and any



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failure to comply with the applicable requirements shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.2 Sensitivity.

4.2.3.2.1 Non-functioning (see 3.4.1) Critical defect - Code No. 05008.

4.2.3.2.1.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 125 primer body and head assemblies shall be selected. If one or more samples fail to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.3.2.1 using equipment in accordance with 4.2.4.

4.2.3.2.1.2 After three consecutive lots have met the criteria of 4.2.3.2.1.1, a sample of 50 primer body and head assemblies shall be selected from each lot. If one or more samples fail to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.3.2.1.

4.2.3.2.2 Functioning (see 3.4.2) Major defect - Code No. 05009.

4.2.3.2.2.1 Beginning with the first lot produced and continuing until three consectuive lots have complied with the applicable requirements specified, 250 samples shall be randomly selected from each lot for this test. If three or more primer body and head assemblies fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of 500 primers shall be tested. If the combined number of defectives in both the first and second sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.2.2. using equipment in accordance with 4.2.4.

4.2.3.2.2.2 After three consecutive lots have met the criteria of 4.2.3.2.2.1 a sample of 125 Primer body and head assemblies shall be selected from each lot for test. If two or more assemblies fail to comply with the requirement, the lot shall be rejected. If only one defective is found, a second sample of 250 assemblies shall be tested. If the combined number of defectives in both the first and second samples is two or more the lot shall be rejected. The test be performed as specified in 4.3.2.2.



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4.2.3.3 Primer assembly functioning (see 3.5) Major defect - Code No. 05010-A sample of 32 primer assemblies shall be selected from each lot. If one or more assemblies fail to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.3.3.

4.2.3.4 Vacuum test - Major defect (see dwg. 8863394 Code No. 05011.-A sample of 32 primer assemblies shall be selected from each lot. If one or more assemblies fail to comply with the requirement specified, the lot shall be rejected. The test shall be performed as specified in 4.3.4 using equipment in accordance with 4.2.4.

4.2.4 Inspection Equipment.-Equipment Tabulation Number ET-8863394 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.- the contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test method and procedures

4.3.1 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.3.2 Sensitivity



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4.3.2.1 Non-functioning.-The sample primer body and head assemblies shall be assembled into the test fixture (see 4.2.4). A steel ball weighing a minimum of 1.92 ounces shall be dropped from a height adjusted to give the required impact force (approximately 1.5 inches).

4.3.2.2 Functioning.-The test shall be performed as specified in 4.3.2.1 except that the weight of the ball shall be 1.96 ounces maximum, and the height shall be adjusted to give the required impact force (approximately 15 inches).

4.3.3 Proving ground.-The primer assembly shall be assembled to the complete round for which the primer was designed and fired from the gun for which the round is standard. Observation shall be made for functioning, hangfires and correct velocity level. Muzzle velocity of each round fired shall be recorded for information.

4.3.4 Vacuum test.-The loaded primer assemblies shall be tested for waterproofness by application of an externally applied vacuum of 10 inches of mercury maintained for a period of 15 seconds, minimum, with the primers submerged under one inch minimum depth of water at 70 plus or minus 3 degrees Fahrenheit. Any primer which fails to meet the applicable requirement shall be classed defective. Visual observation of escaping air into the water from the primer because of inadequate seals will determine non-compliance with the requirements of the applicable drawings.

## 5. PREPARATION FOR DELIVERY

### 5.1 Preservation and Packaging

5.1.1 Level C.-The primers shall be preserved and packaged in accordance with dwg. 8864248.

### 5.2 Packing

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 8864249.

5.3 Marking.-Marking shall be in accordance with dwgs. 8864248 and 8864249.

## 6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.



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b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate



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his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit two copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Hangfire.-A perceptible delay between initiation and detonation.



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6.5 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract.

6.5.1 Primer, body and head assembly

6.5.1.1 Non-functioning (see 3.4.1).-The sampling for the test shall be in accordance with Table I. Any primer, body and head assembly which fails to comply with the applicable requirements shall be classed defective. The test shall be performed as specified in 4.3.2.1, except that the black powder shall be removed from the assembly. The removal of the black powder shall be accomplished in a manner that will not damage the assembly to the extent that the validity of the test results would be questionable. All primer assemblies so tested shall be scrapped.

TABLE I

| Sample | Sample Size | Cumulative Sample Size | Combined Samples Acceptance Number | Combined Samples Rejection Number |
|---|---|---|---|---|
| First | 20 | 20 | * | 2 |
| Second | 20 | 40 | * | 2 |
| Third | 20 | 60 | 0 | 2 |
| Fourth | 20 | 80 | 1 | 3 |
| Fifth | 20 | 100 | 1 | 3 |
| Sixth | 20 | 120 | 1 | 3 |
| Seventh | 20 | 140 | 2 | 3 |

*No acceptance on this sample

6.5.1.2 Functioning (see 3.4.2).-The sampling plan for the test shall be in accordance with Table II. Any primer, body and head assembly which fails to comply with the applicable requirements shall be rejected. The test shall be performed as specified in 4.3.2.2 except that the black powder shall be removed from the assembly. The removal of the black powder shall be accomplished in a manner that will not damage the assembly to the extent that the validity of the test results would be questionable. All primer assemblies so tested shall be scrapped.



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TABLE II

| Sample | Sample Size | Cumulative Sample Size | Combined Samples Acceptance Number | Combined Samples Rejection Number |
|---|---|---|---|---|
| First | 75 | 75 | * | 2 |
| Second | 75 | 150 | 0 | 3 |
| Third | 75 | 225 | 0 | 3 |
| Fourth | 75 | 300 | 1 | 4 |
| Fifth | 75 | 375 | 2 | 5 |
| Sixth | 75 | 450 | 2 | 5 |
| Seventh | 75 | 525 | 4 | 5 |

*No acceptance on first sample

6.5.2 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the primer assembly by any suitable means. **The moisture content of the powder shall be determined in accordance with the methods specified in Specification MIL-P-223. If the moisture content of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

**Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

6.6 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the primers in the cartridges and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lots were under the jurisdiction of either contractor.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-657



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